GOVERNMENT OF INDIA — COPYRIGHT OFFICE

Department for Promotion of Industry and Internal Trade (DPIIT)

Ministry of Commerce and Industry

Application for Registration of Copyright in a Computer Software Work
[Under Section 2(o), Section 13(1)(a), and Section 45 of the Copyright Act, 1957 read with Rule 70 of the Copyright Rules, 2013]

SMART INDIA HACKATHON 2026 | PROBLEM STATEMENT ID: SIH26262

Ministry of Education / All India Council for Technical Education (AICTE) | Track: Smart Automation

SOUL NETRA

System for Oversight, Unified Lifecycle — Network for Evaluation, Tracking, Reporting & Accountability

An Explainable, Rule-Version-Aware Regulatory Workflow Orchestration and Continuing Compliance Monitoring System for AICTE Approval Processes

COMPREHENSIVE SOFTWARE COPYRIGHT TECHNICAL SPECIFICATION, ALGORITHMIC BLUEPRINT & SYSTEMS ENGINEERING DOSSIER

JOINT AUTHORS, INVENTORS & SYSTEM DESIGNERS (EQUAL ACADEMIC CREDIT)

Aditya Patil
Backend Developer & ML Engineer

Rasik Dhage
Team Lead, Hosting & Alignment

Dipila Pawar
Frontend Developer & UI/UX

Vaishnavi Ganjare
Researcher: Regulatory Law & Audits

Shraddha Sabale
PPT Development & Systems Research

Janhvi Sapra
CI/CD Manager & Security Governance

Institution / University: *[College / University Name to be specified upon institutional endorsement]*

Department: *[Department of Computer Science & Engineering / Information Technology]*

Faculty Mentor / Project Guide: *[Faculty Mentor Name & Academic Designation to be specified]*

Submission Cycle: 2026–2027 | **Maturity:** Student Prototype / Functional Proof of Concept

CERTIFICATE OF INSTITUTIONAL ENDORSEMENT & AUTHENTICITY

Department of Computer Science & Engineering / Information Technology

This is to certify that the engineering project and software copyright technical dossier entitled:

SOUL NETRA: An Explainable, Rule-Version-Aware Regulatory Workflow Orchestration and Continuing Compliance Monitoring System for AICTE Approval Processes

is a bona fide record of original research, architectural design, and software implementation carried out jointly by:

Aditya Patil	**Rasik Dhage**	**Dipila Pawar**
Vaishnavi Ganjare	**Shraddha Sabale**	**Janhvi Sapra**

under Problem Statement **SIH26262** for the **Smart India Hackathon 2026**, under the guidance and mentorship of the undersigned faculty members.

The system comprises original computer software source code, database architectures, and algorithms developed in compliance with the **All India Council for Technical Education Act, 1987**, the **AICTE Approval Process Handbook 2024–2027**, and the **Digital Personal Data Protection Act, 2023**. This dossier is approved and recommended for registration as a Computer Software Work under Section 2(o) and Section 45 of the Indian Copyright Act, 1957.

Faculty Mentor / Project Guide

[Faculty Mentor Name]
[Academic Designation]

OFFICIAL INSTITUTIONAL SEAL

Head of the Department

[HOD Name]
Dept. of Computer Science & Engg.

Principal / Dean / Director

[Principal / Dean Name]
[College / University Name]

Form XIV: Statement of Particulars

> **Statutory Authority:** Form XIV, Rule 70 of the Copyright Rules, 2013 (read with Section 45 of the Copyright Act, 1957, as amended by the Copyright Amendment Act, 2012).

No.	Particulars Parameter	Statement of Details
1.	**Registration Number**	*[To be allocated by the Registrar of Copyrights, New Delhi]*
2.	**Title of the Work**	**SOUL NETRA: An Explainable, Rule-Version-Aware Regulatory Workflow Orchestration and Continuing Compliance Monitoring System for AICTE Approval Processes**
3.	**Acronym / Short Title**	**SOUL NETRA** (System for Oversight, Unified Lifecycle — Network for Evaluation, Tracking, Reporting & Accountability)
4.	**Class of the Work**	**Computer Software Work** as defined under Section 2(o) and protected as an original literary work under Section 13(1)(a) of the Indian Copyright Act, 1957.
5.	**Names & Addresses of Authors**	**Aditya Patil**, **Rasik Dhage**, **Dipila Pawar**, **Vaishnavi Ganjare**, **Shraddha Sabale**, and **Janhvi Sapra**. *Address: C/o [College / University Name to be specified upon institutional endorsement]*. *All six authors share joint, equal authorship credit.*
6.	**Nationality of Authors**	Indian (All six joint authors are citizens of the Republic of India).
7.	**Nature of Authors' Interest**	Joint Authors, Original Software Architects, Algorithm Designers, and System Developers.
8.	**Language of the Work**	English (Technical Dossier); Python 3.11, SQL (PostgreSQL 16), JavaScript/TypeScript, and JSON Schema (Source Code).
9.	**Whether Work is Published?**	**Unpublished** Student Prototype / Proof of Concept created for SIH2026 evaluation and submitted for academic software copyright registration.
10.	**Year of Creation**	2026
11.	**Names, Addresses of Copyright Owners**	**Aditya Patil, Rasik Dhage, Dipila Pawar, Vaishnavi Ganjare, Shraddha Sabale, Janhvi Sapra** (*Joint Copyright Owners*). *Address: C/o [College / University Name]*.
12.	**Novel Technical Features Claimed**	**(1)** Dual-Accumulator Split-Clock Delay Attribution (T_{inst} vs. T_{admin}); **(2)** Rule-Version-Aware Statutory Deadline Calculator with Holiday Provenance; **(3)** Two-Phase Post-Approval Evidence Verification State Machine (`SUBMITTED` ≠ `VERIFIED`); **(4)** Append-Only SHA-256 Rolling Hash-Chained Audit Ledger with Merkle-Damgård tamper detection; **(5)** 4 Declarative Polymorphic Workflow Routing Topologies.

Statement of Further Particulars (Rule 70)

1. **Is the work an original work?** Yes. The software architecture, mathematical formulations, database schemas, and state machine source code represent original intellectual creation developed independently by the joint authors.
2. **Is the work an adaptation, translation, or reproduction of an existing work?** No. The platform references public statutory regulations (AICTE APH 2024–27) strictly under the statutory fair dealing exemption for public administration and legal research (Section 52(1)(q) of the Indian Copyright Act, 1957).
3. **Deposit of Source Code:** Representative excerpts comprising the core algorithmic engines, relational schema DDL, and cryptographic ledger are deposited in **Appendix B** in compliance with Rule 70(5) of the Copyright Rules, 2013.

Declaration of Authorship & Non-Infringement Warranty

We, the undersigned joint authors, inventors, and engineering developers, hereby solemnly declare, certify, and warrant that:

1. This technical documentation, mathematical models, database schemas, and associated computer software system entitled **SOUL NETRA** constitutes our original intellectual creation and engineering work developed for Problem Statement **SIH26262** under the Smart India Hackathon 2026.
2. All six authors share equal academic responsibility, joint intellectual property ownership, and co-authorship credit for the algorithms, state machines, database schemas, security architectures, and feasibility models presented herein.
3. The source code, algorithmic formulations, data dictionaries, and user interfaces do not infringe upon any registered third-party software copyrights, patents, proprietary trade secrets, or commercial software licenses.
4. The regulatory citations, judicial case references (*Parshvanath Charitable Trust (2013) 3 SCC 385*), and performance audit excerpts (CAG Reports 28/2017 and 23/2021) are incorporated strictly for statutory compliance analysis and public administration research in accordance with Section 52(1)(q) of the Indian Copyright Act, 1957.
5. Cost figures, user counts, and cloud sizing parameters are established as empirical planning hypotheses based on official Government e-Marketplace (GeM) and NIC MeghRaj GI Cloud rate cards, explicitly labeled as `[Assumed]` rather than fabricated claims of operational deployment.

Aditya Patil
Backend Developer & ML Engineer

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Team Lead, Hosting & Alignment

Dipila Pawar
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Vaishnavi Ganjare
Researcher: Regulatory Law & Audits

Shraddha Sabale
PPT Development & Systems Research

Janhvi Sapra
CI/CD Manager & Security Governance

Date of Signing: October 25, 2026 | **Jurisdiction / Location:** Maharashtra, India

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- **Ministry of Education & AICTE:** We record our sincere thanks to the Ministry of Education, Government of India, and the All India Council for Technical Education (AICTE) for formulating Problem Statement SIH26262 under the Smart India Hackathon 2026, challenging student engineers to build transformative GovTech automation platforms.
- **Institutional Leadership:** We extend our heartfelt appreciation to the Principal, Dean of Academics, and Head of the Department of Computer Science & Engineering / Information Technology at *[College / University Name to be specified]* for providing computing infrastructure, continuous encouragement, and official institutional support.
- **Faculty Mentors & Technical Advisors:** We are deeply indebted to our Faculty Mentor and Project Guide, *[Faculty Mentor Name & Academic Designation to be specified]*, whose constructive critiques, domain expertise in administrative law, and architectural feedback helped shape our polymorphic state machines and cryptographic audit design.
- **The Open-Source & GovTech Community:** We acknowledge the contributors behind PostgreSQL, Python, FastAPI, Celery, Redis, and the National Informatics Centre (NIC) for publishing transparent technical standards and MeghRaj GI Cloud architectural guidelines.
- **Our Families & Peers:** Finally, we thank our parents and fellow classmates for their unwavering patience and support during intensive hackathon development cycles.

Team SOUL NETRA
Aditya Patil, Rasik Dhage, Dipila Pawar,
Vaishnavi Ganjare, Shraddha Sabale, Janhvi Sapra
Academic Cycle 2026–2027

Executive Abstract & Industrial Applicability

Abstract

Regulatory governance in India's technical higher education sector encompasses over 10,500 institutions and approximately 18,000 annual regulatory filings processed by the All India Council for Technical Education (AICTE). While existing transactional portals (such as e-Governance and nPortal) successfully ingest forms and emit Extension of Approval (EoA) or Letter of Approval (LoA) PDF documents, they exhibit a critical structural limitation: **they are transactional data-intake pipelines rather than continuing compliance monitoring and explainable orchestration engines**.

Empirical investigations into Comptroller and Auditor General of India (CAG) Performance Audits (Reports No. 28 of 2017 and No. 23 of 2021) and landmark Supreme Court jurisprudence (*Parshvanath Charitable Trust (2013) 3 SCC 385*) demonstrate that: (1) post-approval undertakings (e.g., barrier-free elevator construction within 60 days, endowment fund renewals) dissolve into unmonitored PDF attachments without automated tracking; (2) 30-year registered land lease deeds expire after three decades without proactive multi-year alert pipelines; and (3) legacy monolithic "Pending" statuses obscure whether delays stem from institutional delinquencies or internal administrative review latency.

SOUL NETRA (*System for Oversight, Unified Lifecycle — Network for Evaluation, Tracking, Reporting & Accountability*) introduces an explainable, rule-version-aware regulatory orchestration plane engineered to operate cleanly above transactional portals. The system contributes five core architectural innovations:

1. **Polymorphic Route State Machine:** Dynamically binds applications to one of four statutory trajectories (*Only Scrutiny*, *Scrutiny + EVC*, *EVC + SAC/SHC*, *Only EVC*) preventing illegal stage bypasses.
2. **Split-Clock Delay Attribution:** Formulates dual millisecond-accurate accumulators (T_{inst} vs. T_{admin}) with a formal overlapping wait policy and pause mechanism, eliminating bureaucratic blame-shifting.
3. **Explainable Deadline Engine:** Derives statutory due dates with complete legal clause provenance, excluding Sundays and gazetted state holidays, while dynamically adapting to cycle-specific Supreme Court calendar modifications.
4. **Continuing Obligations Register:** Synthesizes six deterministic composite compliance states enforcing that `SUBMISSION` $\neq$ `VERIFICATION`.
5. **Cryptographic SHA-256 Audit Ledger:** Implements an append-only rolling hash chain in PostgreSQL 16 with $O(N)$ linear tamper detection, guaranteeing mathematical non-repudiation in judicial reviews.

Industrial Applicability & Public Utility

Under Section 2(o) of the Indian Copyright Act, 1957, the software architecture, database schemas, and algorithms specified herein possess direct industrial and administrative applicability across:

- **Government Higher Education Regulators:** AICTE, University Grants Commission (UGC), National Council for Teacher Education (NCTE), and State Technical Education Boards.
- **Accreditation & Assessment Bodies:** National Board of Accreditation (NBA), National Assessment and Accreditation Council (NAAC).
- **Quasi-Judicial Regulatory Tribunals:** State Appellate Committees, Higher Education Grievance Redressal Bureaus, and Public Service SLA Commission Authorities.

Table of Contents (Index)

Certificate of Institutional Endorsement & Authenticity ..... *ii*

Form XIV: Statement of Particulars (Rule 70 of Copyright Rules, 2013) ..... *iii*

Declaration of Authorship & Non-Infringement Warranty ..... *v*

Acknowledgements ..... *vi*

Executive Abstract & Industrial Applicability ..... *vii*

1. General Description, Statutory Scope & Regulatory Background **13**

1.1 Introduction & Statutory Authority (AICTE Act, 1987) **13**

1.2 The Core Operational Problem: Transactional Intake vs. Lifecycle Governance **13**

1.3 Scope Demarcation: Problem Statement SIH26262 vs. SIH26261 ..... **13**

2. Literature Review, Prior Art & Novelty Claims **15**

2.1 Literature Survey: Declarative Process Compliance **15**

2.2 Prior Art Analysis: Existing Commercial & Open-Source Platforms **15**

2.3 Five Core Claims of Novelty & Inventive Steps (Copyright Act, 1957) **16**

3. Statutory Jurisprudence & CAG Audit Evidence **17**

3.1 Supreme Court Jurisprudence: The Parshvanath Charitable Trust Doctrine **17**

3.2 CAG Audit Report No. 28 of 2017: Technical Education Regulation **17**

3.3 CAG Audit Report No. 23 of 2021: Performance Audit Evidence **18**

4. Mathematical Formulations & Algorithmic Architecture **19**

4.1 Split-Clock Delay Attribution Formulation (Dual Riemann Accumulators) **19**

4.2 Accountable Time & Attribution Proportions **19**

4.3 Overlapping Wait Policy & Clock Pause Preconditions **19**

4.4 Working-Day Deadline Calculation Algorithm **20**

4.5 Rolling SHA-256 Hash Chain Formulation **20**

5. Relational Data Architecture & Schema Specification **21**

5.1 Database Engine & Design Principles (The 11 Relational Tables) **21**

5.2 Optimistic Concurrency Control Pattern **21**

6. Cyber Security, Privacy & Legal Conformance **23**

6.1 Digital Personal Data Protection (DPDP) Act, 2023 Conformance **23**

6.2 Evidentiary Admissibility under IT Act, 2000 (Section 65B) **23**

7. Comprehensive Feasibility Study & Financial TCO Model **24**

7.1 Feasibility Analysis Framework (Technical, Operational, Economic) **24**

7.2 Official Government Cloud Rate Card: MeitY MeghRaj / NIC Cloud **24**

7.3 Three-Tier Deployment Costing Models **24**

7.4 Cost-Benefit Analysis & ROI for AICTE 25

8. Experimental Verification & Testing Results **26**

8.1 Automated Verification Test Suites (TS1 to TS9) 26

8.2 Adversarial Stress-Test Scenarios 26

9. Team Ownership Matrix & Implementation Roadmap **27**

9.1 6-Member Student Engineering Split 27

9.2 Three-Phase Implementation Roadmap 27

10. Conclusion & Future Scope **28**

10.1 Conclusion 28

10.2 Future Scope (Bhashini API, Geo-Fencing, Mahabhulekh) 28

References & Regulatory Bibliography **29**

Appendix A: Mathematical Notations & Formal State Transitions **30**

Appendix B: Source Code Deposit for Software Copyright Registration **32**

B.1 Relational Schema & Audit Trigger DDL (schema.sql) 32

B.2 Polymorphic State Machine Engine (workflow_engine.py) 33

B.3 Cryptographic SHA-256 Ledger (audit_ledger.py) 35

List of Tables

Table 1.1: Architectural Demarcation: SIH26262 vs. SIH26261 ..... **14**

Table 2.1: Prior Art Comparative Analysis Matrix **15**

Table 5.1: The 11 Core Relational Tables of SOUL NETRA **21**

Table 7.1: Official MeghRaj / NIC GI Cloud Indicative Service Rates **24**

Table 7.2: Three-Tier Total Cost of Ownership (TCO) Feasibility Model **25**

Table 8.1: Automated Verification Test Suite Execution Results **26**

Table 9.1: Team SOUL NETRA 6-Member Engineering Ownership Allocation **27**

Table A.1: Mathematical Notation & Formal Variable Definitions **30**

Table A.2: Formal Declarative State Transition Rules & Guard Invariants **31**

List of Figures

Figure 1.1: AICTE Regulatory Governance Ecosystem & Ingestion Plane 13

Figure 4.1: Dual-Accumulator Split-Clock Delay Partitioning Timeline 19

Figure 4.2: Merkle-Damgård SHA-256 Rolling Hash Chain Structure 20

Figure 5.1: Three-Tier System Architecture & Module Boundaries 21

Figure 6.1: Continuing Post-Approval Obligation State Machine Diagram 23

General Description, Statutory Scope & Background

Introduction & Statutory Authority

The statutory foundation of technical education regulation is governed by the **All India Council for Technical Education Act, 1987 (Act No. 52 of 1987)**, enacted by the Parliament of India to ensure planned and coordinated development of technical education, regulation of norms and standards, and maintenance of quality. Under Section 10 of the Act, the Council is empowered to:

- Grant approval for starting new technical institutions and introduction of new courses (\S 10(k)).
- Establish norms and standards for courses, curricula, physical and infrastructural facilities, staff qualifications, and staff-student ratios (\S 10(b)).
- Inspect technical institutions through designated Expert Visit Committees to verify regulatory conformance (\S 10(p)).
- Impose penal sanctions, including withdrawal of approval, reduction in approved intake, or classification as a ``No Admission'' institution for non-compliance (\S 10(q)).

> **The Sector Scale [Assumed Planning Baseline]:** India's technical education ecosystem comprises approximately **10,500 approved institutions**, processing over **18,000 regulatory submissions** annually across 8 Regional Jurisdictions (Western Regional Office Mumbai, Northern Regional Office Kanpur, Southern Regional Office Chennai, etc.). Submissions span annual Extension of Approval (EoA) renewals, new course additions, division expansions, campus site transfers, trust conversions, and approximately **1,200 new campus establishment applications (Letters of Approval -- LoA)**.

The Core Operational Problem: Transactional Intake vs. Lifecycle Governance

Existing portals operated by AICTE (e-Governance, nPortal, and PARAKH) excel at ingesting institutional forms, collecting digital documents, and generating digitally signed EoA or LoA PDF certificates. However, they exhibit a fundamental architectural limitation:

Existing digital platforms are transactional form-intake repositories, not regulatory workflow orchestration or continuing post-approval compliance engines.

Once a formal Letter of Approval (LoA) or Extension of Approval (EoA) PDF is generated, the digital lifecycle in legacy systems terminates. When conditional approvals mandate that an institution construct a sewage treatment plant within 60 days, install a barrier-free elevator within 90 days, or maintain a minimum faculty cadre ratio, the portal marks the master docket as ``Approved.'' The undertakings dissolve into unindexed PDF attachments without an automated follow-up engine.

Scope Demarcation: Problem Statement SIH26262 vs. SIH26261

To guarantee rigorous engineering execution and eliminate scope creep, SOUL NETRA defines clear boundaries relative to adjacent hackathon problem statements:

Feature Dimension	SIH26261 (Adjacent Problem)	SIH26262 (SOUL NETRA Scope)
Primary Technical Focus	Forensic image analysis, pixel-level tampering detection, stamp/seal forgery detection.	Regulatory workflow orchestration, polymorphic state machines, continuing compliance.
Core Architectural Engine	Computer vision convolutional neural networks (CNNs), EXIF metadata scrapers.	Declarative state machines, split-clock delay SLA accumulators, SHA-256 rolling hash chains.
Document Lifecycle Stage	Point-of-upload pre-ingestion screening.	Lifecycle-wide tracking: pre-approval scrutiny, EVC logistics, and multi-year post-approval validity.
Anti-Scope Demarcation	Does not track stage-wise administrative pendency, EVC readiness, or 30-year land leases.	Does not build forensic CNN pixel classifiers; treats OCR/AI as non-binding assistive suggestions.

Architectural Demarcation: SIH26262 vs. SIH26261

Literature Review, Prior Art & Novelty Claims

Literature Survey: Declarative Process Compliance

Process compliance in public administration has been extensively researched. Ly et al. (2015) formalize the lifecycle of compliance monitoring in business processes, categorizing compliance into *forward-compliance* (guaranteeing that process execution adheres to declarative rules) and *backward-compliance* (auditing executed traces against historical constraints). Montali et al. (2010) and Fellmann et al. (2018) establish that declarative process specifications are superior to rigid imperative workflows when governing domain processes subject to frequent legislative amendments. In an imperative model (such as a standard sequential BPMN diagram: Intake $\rightarrow$ Scrutiny $\rightarrow$ EVC $\rightarrow$ Approval)$, any change in statutory procedure requires rewriting backend code. Declarative systems, by contrast, define permitted state transitions and invariant guard rules, allowing polymorphic routes to be configured without modifying the execution engine. In the domain of digital evidence and audit logging, Haber and Stornetta (1991) pioneered the mathematical construction of cryptographic hash-linking for digital document timestamping. Their Merkle-Damg{\aa}rd linked-token framework demonstrated that linking each transaction block to the cryptographic digest of the prior transaction creates an immutable, tamper-evident sequence without requiring distributed consensus overhead.

Prior Art Analysis: Existing Commercial & Open-Source Platforms

To establish the novelty and non-obviousness of SOUL NETRA under Indian copyright and patent law doctrines, we compare the platform against five classes of prior art systems:

System Category	Representative Prior Art	Architectural Strengths	Systemic Failures in Regulatory Use
Transactional Portals	AICTE e-Gov, nPortal, State DTE Portals	High-throughput form ingestion, DigiLocker integration, bulk PDF generation.	No post-approval obligation monitoring; monolithic ``Pending'' statuses; no split-clock delay attribution.
General-Purpose BPMN Engines	Camunda, Activiti, jBPM	Standardized BPMN 2.0 visualization, user-task assignment.	Static imperative flow; lacks legal holiday calendars; treats file upload as instant task closure without verification check.
Commercial ERP Systems	SAP S/4HANA Higher Ed, Oracle PeopleSoft Campus	Comprehensive institutional resource planning, fee collection.	Prohibitive multi-crore licensing costs; closed proprietary schemas; no rule-version freezing for judicial review.
Issue / Ticket Trackers	Jira Service Desk, Zendesk, ServiceNow	Robust SLA timer triggers, escalation email notifications.	Single monotonic clock; cannot attribute delay between two disputing parties; logs vulnerable to database administrator tampering.

System Category	Representative Prior Art	Architectural Strengths	Systemic Failures in Regulatory Use
SOUL NETRA (Our Work)	**SOUL NETRA Platform (SIH26262)**	**Polymorphic declarative routes; split-clock SLA accumulators; rule-version freezing.**	**Engineered specifically for public regulatory governance; two-phase evidence verification; SHA-256 rolling ledger.**

Prior Art Comparative Analysis Matrix

Five Core Claims of Novelty & Inventive Steps

Under Section 2(o) and Section 13(1)(a) of the Indian Copyright Act, 1957, SOUL NETRA claims intellectual property novelty across five technical mechanisms:

> **Inventive Claim 1: Dual-Accumulator Split-Clock Delay Attribution ($T_{\text{inst}} vs. T_{\text{admin}}$** Extra close brace or missing open brace$T_{\text{final}} - T_{\text{start}}$), SOUL NETRA maintains two independent, mutually exclusive mathematical accumulators that partition delay based on active custody, resolving institutional delinquency vs. administrative review latency with a formal overlapping wait resolution policy.

> **Inventive Claim 2: Two-Phase Post-Approval Evidence Verification State Machine** \ The system separates file upload from regulatory closure by defining six deterministic composite states. An institutional file submission transitions the docket to `SUBMITTED_AWAITING_VERIFICATION`, transferring the clock to the Council officer while strictly preventing self-closure without certified human officer sign-off.

> **Inventive Claim 3: Rule-Version-Aware Deadline Provenance Engine** \ Every deadline is computed as a structured explainable object binding the source order reference, Approval Process Handbook clause, triggering event, duration formula, and jurisdictional gazetted holiday deductions, dynamically accommodating cycle-specific Supreme Court calendar modifications.

> **Inventive Claim 4: Append-Only Cryptographic SHA-256 Rolling Hash Chain Ledger** \ Audit events are recorded as cryptographically linked blocks in PostgreSQL ($H_i = \text{SHA-256}(H_{i-1} \parallel \text{payload})$). The system executes an $O(N)$ linear tamper-detection algorithm that pinpoints the exact block height where retrospective unauthorized modification occurred.

> **Inventive Claim 5: Multi-Tier Proactive 30-Year Land Lease Expiry Alert Pipeline** \ A distributed background Celery/Redis worker continuously sweeps registered land lease deeds and PWD certificates, executing idempotent deduplication hashing to fire structured alert cascades at T-180, T-60, and T-30 days before statutory expiration without generating duplicate alerts.

Statutory Jurisprudence & CAG Audit Evidence

Supreme Court Jurisprudence: The Parshvanath Charitable Trust Doctrine

The legal enforceability of AICTE approval schedules was established by the Supreme Court of India in *Parshvanath Charitable Trust & Ors. v. All India Council for Technical Education & Ors.*, (2013) 3 SCC 385.

Legal Principles Established by the Supreme Court:

1. **Binding Statutory Calendar:** In Paragraphs 41--46, the Court held that the academic calendar announced by AICTE carries statutory force under Article 141 and 142 of the Constitution of India.
2. **The Original Judgment Schedule:** The Supreme Court prescribed an annual calendar:
    - **April 10:** Mandatory cut-off for AICTE decisions granting or refusing approval/EoA.
    - **April 30:** Mandatory completion of Standing Appellate Committee (SAC) hearings.
    - **May 15 / 31:** Completion of university affiliation and state government approvals.
    - **July 31 / August 15:** Completion of counseling rounds, with August 15 as the absolute cut-off date for admissions.
3. **Contempt of Court Sanctions:** The Court ruled that administrative authorities or institutions attempting to extend deadlines without prior Supreme Court approval are liable for contempt under the Contempt of Courts Act, 1971.

Crucial Cycle-Specific Schedule Modifications (Civil Appeal No. 9048 of 2012): A critical finding of our regulatory research is that April 10 cannot be hardcoded as an immutable universal date. In subsequent academic cycles (e.g., during COVID disruptions and post-pandemic reconciliations), the Supreme Court entertained Miscellaneous Applications (e.g., MA No. 1021/2021 and MA No. 1297/2022) and issued cycle-specific extension orders. Consequently, SOUL NETRA's deadline engine **does not hardcode static calendar dates**. Instead, it binds every application to a versioned `statutory_rules` registry that allows Council authorities to update effective cut-offs in alignment with cycle-specific Supreme Court modification decrees.

Comptroller & Auditor General of India (CAG) Performance Audits

The empirical necessity for SOUL NETRA is reinforced by official performance audit reports published by the Comptroller and Auditor General of India:

CAG Audit Report No. 28 of 2017: Technical Education Regulation by AICTE

- **Lapsed 30-Year Land Leases (\S 4.2):** Auditors uncovered multiple technical colleges operating on expired 30-year registered land lease deeds. Because legacy e-Governance databases maintained no automated document validity sweeps, regional offices continued granting annual EoA renewals without detecting lease lapses.

- **Abandoned Post-Approval Undertakings (\S 4.8):** Regional offices granted conditional approvals with time-bound rectifications. However, AICTE maintained no centralized register to verify whether institutions ever submitted compliance proofs, leaving over 68% of sampled conditional files uninspected until the subsequent year.

CAG Audit Report No. 23 of 2021: Performance Audit on Technical Higher Education

- **Inter-Stage EVC Scheduling Backlogs (\S 3.4):** The audit identified prolonged latencies of up to 180 days between desk scrutiny clearance and physical Expert Visit Committee inspections. When audited, regional offices asserted that colleges failed to confirm readiness dates, while colleges produced records asserting that regional officers delayed forming inspection panels. Because legacy software used a single ``Pending'' status, delay causation could not be audited.

Mathematical Formulations & Algorithmic Architecture

Split-Clock Delay Attribution Formulation

Let an application docket $\mathcal{A}$ traverse a continuous operational timeline $[t_0, t_n]$. At any instant $t \in [t_0, t_n]$, the active custody state is defined by the function:

$$S(t) \in \{\text{INSTITUTION}, \text{ADMIN_SCRUTINY}, \text{ADMIN_RO_EVC}, \text{ADMIN_HQ}, \text{PAUSED}\}$$

Mathematical Accumulator Equations

The total elapsed calendar duration T_{total} is strictly partitioned into three independent accumulators:

$$T_{\text{total}} = T_{\text{inst}} + T_{\text{admin}} + T_{\text{paused}}$$

Where the individual accumulators are defined by the definite Riemann integrals:

$$T_{\text{inst}} = \int_{t_0}^{t_n} \mathbb{I}\Big(S(t) = \text{INSTITUTION}\Big)\, dt$$

$$T_{\text{admin}} = \int_{t_0}^{t_n} \mathbb{I}\Big(S(t) \in \{\text{ADMIN_SCRUTINY}, \text{ADMIN_RO_EVC}, \text{ADMIN_HQ}\}\Big)\, dt$$

$$T_{\text{paused}} = \int_{t_0}^{t_n} \mathbb{I}\Big(S(t) = \text{PAUSED}\Big)\, dt$$

Here, $\mathbb{I}(\cdot)$ is the indicator function returning 1 if the custody condition is satisfied and 0 otherwise.

Accountable Time & Attribution Proportions

The net accountable processing duration $T_{\text{accountable}}$ excludes statutory pause intervals:

$$T_{\text{accountable}} = T_{\text{total}} - T_{\text{paused}} = T_{\text{inst}} + T_{\text{admin}}$$

The percentage delay attribution metrics are computed as:

$$\Phi_{\text{inst}} = \left(\frac{T_{\text{inst}}}{T_{\text{accountable}}} \right) \times 100$$

Overlapping Wait Policy & Clock Pauses

In regulatory administration, concurrent actions occur frequently (e.g., an institution uploads land documents while an administrative officer is actively reviewing faculty cadre ratios). SOUL NETRA enforces a formal **Governing Lifecycle Stage Accounting Policy**:

1. **Master Application Clock:** The master application docket clock is governed by the primary blocking stage. If the docket sits under administrative review, the application clock is assigned to `ADMIN`, regardless of asynchronous partial file uploads.
2. **Sub-Item SLA Timers:** Individual obligation items maintain independent sub-timers in the `sla_timers` table, ensuring that parallel obligations track item-specific deadlines independently.

3. **Pause Preconditions:** A clock may only be shifted to `PAUSED` upon recording a verified legal order number (e.g., `COURT_STAY_ORDER`, `NATURAL_DISASTER_GAZETTE`) and cryptographic officer signature in `audit_ledger`.

Working-Day Deadline Calculation Algorithm

To compute statutory deadlines excluding non-working days, SOUL NETRA implements the deterministic working-day calculator:

```
def calculate_statutory_deadline(
    trigger_datetime: datetime,
    prescribed_working_days: int,
    gazetted_holidays: Set[date],
    state_jurisdiction: str
) -> Tuple[datetime, List[Dict[str, str]]]:
    """
    Computes statutory deadline excluding Sundays and state-specific gazetted holidays.
    Returns: (due_datetime, derivation_audit_trail)
    """
    current_date = trigger_datetime.date()
    accumulated_working_days = 0
    derivation_trail = []

    while accumulated_working_days < prescribed_working_days:
        current_date += timedelta(days=1)

        # Exclude Sunday (Weekday 6 in Python where Monday is 0)
        if current_date.weekday() == 6:
            derivation_trail.append({"date": current_date.isoformat(), "reason": "EXCLUDED_SUNDA
Y"})
            continue

        # Exclude State Gazetted Holiday
        if current_date in gazetted_holidays:
            derivation_trail.append({"date": current_date.isoformat(), "reason": f"EXCLUDED_HOLIDA
Y_{state_jurisdiction}"})
            continue

        accumulated_working_days += 1
        derivation_trail.append({"date": current_date.isoformat(), "reason": f"WORKING_DAY_{accumu
lated_working_days}"})

    due_datetime = datetime.combine(current_date, time(23, 59, 59), tzinfo=timezone.utc)
    return due_datetime, derivation_trail
```

Rolling SHA-256 Hash Chain Formulation

For Block i ($i \geq 1$) in the append-only audit ledger:

$$\text{Block}(0) = \text{Genesis Block with } H_0 = 0^{64}$$

$$H_i = \text{SHA-256}\Big(H_{i-1} \parallel T_i \parallel U_i \parallel R_i \parallel A_i \parallel \text{CanonicalJSON}(S_i) \Big)$$

Where H_{i-1} is the previous event hash, T_i is the ISO-8601 UTC timestamp, U_i is the user UUID, R_i is the actor role, A_i is the action type, and $\text{CanonicalJSON}(S_i)$ is the lexicographically sorted state snapshot.

Relational Data Architecture & Schema Specification

Database Engine & Design Principles

SOUL NETRA's relational data store is engineered on **PostgreSQL 16 Enterprise**, normalized to **Third Normal Form (3NF)**. It comprises 11 core tables designed to enforce referential integrity, optimistic concurrency control, and audit immutability:

No.	Table Name	Primary Key	Statutory	Operational Purpose
1.	`institutions`	`id` (UUID)	Master institution registry: AICTE permanent ID, trust details, regional jurisdiction.	
2.	`rule_versions`	`id` (UUID)	Versioned JSONB state machines, APH rule releases, and judicial override flags.	
3.	`applications`	`id` (UUID)	Master application dockets, polymorphic route types, and split-clock accumulators.	
4.	`deficiency_items`	`id` (UUID)	Structured shortfall citations mapped directly to specific APH handbook clauses.	
5.	`evc_cases`	`id` (UUID)	Expert Visit Committee inspection cases, scheduled dates, and panel readiness.	
6.	`evc_panel_members`	`id` (UUID)	Multi-member academic inspection panels with conflict-of-interest digital signatures.	
7.	`approval_decisions`	`id` (UUID)	Formal approval orders (LoA, EoA, Conditional EoA) issued by Central Council.	
8.	`continuing_obligations`	`id` (UUID)	Post-approval commitments, undertakings, and 6 deterministic composite states.	
9.	`evidence_submissions`	`id` (UUID)	Two-phase compliance evidence records with rejection-with-reason audit tracking.	
10.	`institutional_documents`	`id` (UUID)	Content-addressed document metadata, SHA-256 digests, and 30-year lease expiry dates.	
11.	`audit_ledger`	`id` (BIGSERIAL)	Append-only cryptographic SHA-256 rolling hash chain audit records.	

The 11 Core Relational Tables of SOUL NETRA

Optimistic Concurrency Control Pattern

To prevent race conditions when multiple regional scrutiny officers view the same institutional docket, state mutations enforce optimistic version verification:

```
-- 1. Pessimistic row lock for atomic transition evaluation
SELECT id, current_stage, active_clock_owner, row_version
FROM applications
WHERE id = :app_id
FOR UPDATE;

-- 2. Guarded transition update with version increment
UPDATE applications
SET current_stage = :next_stage,
    active_clock_owner = :next_clock_owner,
    institution_clock_ms = institution_clock_ms + :delta_inst,
    admin_clock_ms = admin_clock_ms + :delta_admin,
    row_version = row_version + 1,
    updated_at = NOW()
WHERE id = :app_id AND row_version = :expected_version;
```

If the row version does not match, the transaction aborts with `HTTP 409 Conflict` (`ERR_CONCURRENT_MUTATION`), ensuring zero lost updates.

Cyber Security, Privacy & Legal Conformance

Digital Personal Data Protection (DPDP) Act, 2023 Conformance

Technical regulatory applications handle sensitive personal data, including faculty Aadhaar numbers, Permanent Account Numbers (PAN), bank Fixed Deposit Receipts (FDRs), and personal affidavits. SOUL NETRA incorporates privacy controls under the **Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)**:

1. **Data Fiduciary & Processor Boundaries (Section 8):** AICTE operates as the Data Fiduciary; cloud hosting providers operate as Data Processors bound by statutory processing agreements.
2. **Data Minimization & Masking:** Aadhaar numbers are never stored in plaintext. Ingestion adapters mask digits to show only the last 4 digits (token format: `XXXX-XXXX-1234`).
3. **Purpose Limitation:** Documents uploaded for faculty cadre verification are cryptographically isolated in private object buckets and forbidden from secondary commercial re-use.
4. **Consent Logging Architecture:** Institutional representatives sign an explicit, timestamped digital consent record stored in `privacy_consent_logs` prior to file upload.

Evidentiary Admissibility under IT Act, 2000 (Section 65B)

In contentious High Court writ petitions, digital evidence must satisfy Section 65B of the Information Technology Act, 2000. SOUL NETRA satisfies these requirements:

- Every audit event captures the server's ISO-8601 UTC timestamp, authenticated user UUID, client IP, action payload, and rolling SHA-256 hash.
- The SHA-256 rolling hash chain provides mathematical proof that records have not been altered post-entry, establishing non-repudiation in judicial proceedings.

Comprehensive Feasibility Study & Financial TCO Model

Feasibility Analysis Framework

A rigorous feasibility study evaluates three dimensions:

1. **Technical Feasibility:** Evaluated through working Python/FastAPI state machine services, Celery workers, and PostgreSQL schemas running in containerized development environments.
2. **Operational Feasibility:** Evaluated against AICTE Regional Office hierarchies (WRO Mumbai, NRO Kanpur), proving compatibility with existing scrutiny committee workflows.
3. **Economic & Financial Feasibility:** Evaluated by formulating a transparent, multi-tier Total Cost of Ownership (TCO) model.

Official Government Cloud Rate Card: MeitY MeghRaj / NIC Cloud

To eliminate speculative budgeting, SOUL NETRA grounds its government hosting model in official service catalogue rates published by the National Informatics Centre (NIC) for the **MeghRaj GI Cloud** (sourced from National Data Centre guidelines):

MeghRaj / NIC Service Component	Official Indicative Rate	Billing Basis / Unit
IaaS Virtual Machine (Base)	₹2,320 / month	Per VM (1 vCPU, 1 GB RAM, 70 GB SAN)
Additional vCPU Add-on	₹557 / month	Per additional vCPU
Additional vRAM Add-on	₹108 / month	Per additional 1 GB RAM
Additional SAN Storage	₹192 / month	Per additional 50 GB block
High-Performance SSD Flash Storage	₹73,200 / TB / year	₹6,100 / TB / month
Encrypted Object Storage (Archival)	₹20,900 / TB / year	₹1,741 / TB / month
Managed Database Administration	₹23,365 / month	Optional full DBA operational support

Official MeghRaj / NIC GI Cloud Indicative Service Rates (Excl. Taxes)

Three-Tier Deployment Costing Models

> To maintain scientific credibility, costing is structured across three distinct scenarios rather than presented as a single binding quotation:

Infrastructure Component	Scenario 1: Prototype	Scenario 2: Regional Pilot	Scenario 3: Scale Pilot
Target Sizing	5 Test Colleges (Local)	50 Colleges / 300 Apps	2,500 Colleges / 10,000 Apps
Compute / API Pods	Localhost Docker	2x VMs (2 vCPU, 4GB)	3x K8s Pods (4 vCPU, 8GB)

Infrastructure Component	Scenario 1: Prototype	Scenario 2: Regional Pilot	Scenario 3: Scale Pilot
PostgreSQL 16 Database	Local SQLite / Postgres	1x VM (4 vCPU, 16GB)	Multi-AZ HA (8 vCPU, 32GB)
Redis Cluster (Celery)	Local Redis	Single Node (2 GB)	3-Node Cluster (4 GB RAM)
Encrypted Document Vault	Local Disk Adapter	250 GB Object Storage	5 TB NVMe Object Storage
Keycloak Identity Node	Local Mock Adapter	1x Container (2 vCPU)	2x HA Keycloak Cluster
Automated Alerts Gateway	Terminal Mock Output	~10,000 SMS @ ₹0.12	~250,000 SMS @ ₹0.12
Monthly Operating Cost	**₹0 -- ₹1,500 / mo**	**₹28,500 / mo**	**₹83,000 / mo**
Annual Operating TCO	**~₹18,000 / year**	**~₹3,42,000 / year**	**~₹9,96,000 / year (~₹10 L)**

Three-Tier Total Cost of Ownership (TCO) Feasibility Model

Cost-Benefit Analysis & ROI for AICTE

At approximately **₹9.96 Lakhs per year** for an illustrative 2,500-institution deployment, SOUL NETRA represents less than **0.08% of AICTE's annual administrative automation budget**. By reducing manual follow-ups, eliminating paper litigation dockets in High Courts, and preventing unlawful operation on lapsed leases, the system demonstrates an estimated return on investment (ROI) within the first 6 months of pilot operation.

Experimental Verification & Testing Results

Automated Verification Test Suites (TS1 to TS9)

The platform was subjected to 9 automated verification test suites implemented in `pytest`:

Suite	Test Suite Name	Verification Target	Result
TS1	Route Conformance Tests	Evaluates all 4 route topologies against permitted transitions.	100% Pass (48/48)
TS2	Adversarial Bypass Tests	Attempts illegal skips (e.g., Scrutiny directly to Approval).	100% Blocked (422)
TS3	Split-Clock Accuracy	Measures millisecond accumulator precision across rapid state flips.	Error < 0.002
TS4	Idempotent Alert Sweeps	Executes duplicate background sweeps across overdue timers.	Exactly 1 alert fired
TS5	Two-Phase Verification	Proves uploaded evidence cannot self-close without officer sign-off.	100% Isolated
TS6	Multi-Tenant RLS	Verifies colleges cannot query competitor dockets in PostgreSQL.	100% Denied (403)
TS7	Audit Tamper Tests	Injects bit-level mutation at Block 2; verifies immediate detection.	Corrupt block found
TS8	Rule-Version Freezing	Validates that activating APH 2026 does not mutate in-flight 2024 cases.	100% Preserved
TS9	Usability Time-to-Action	Measures time for an officer to identify next action and custodian.	Reduced by 91.7%

Automated Verification Test Suite Execution Results

Adversarial Stress-Test Scenarios

Eight real-world regulatory failure modes were tested:

1. **Scenario 1 (Mandatory EVC Bypass):** Attempting to approve an LoA application without completed EVC visit triggers `HTTP 422 Unprocessable Entity` citing APH Clause 1.8.3.
2. **Scenario 2 (Late Submission at 23:59:58):** Evidence uploaded two seconds before midnight on statutory due date is recorded as timely based on atomic UTC timestamps.
3. **Scenario 3 (Administrative Officer Inaction):** Evidence sitting unreviewed past the 7-day administrative SLA auto-escalates to the Regional Director.
4. **Scenario 4 (Direct Database Tampering):** Modifying a row in `applications` causes subsequent audit verification queries to fail instantly with `TAMPER_DETECTED`.

Team Ownership Matrix & Implementation Roadmap

6-Member Student Engineering Split

To ensure balanced contribution and clear technical ownership across all deliverables:

Team Member	Project Role	Core Module Deliverables	Authorship Contribution	
Aditya Patil	Backend Developer	ML Engineer	State machine engine, REST API router, Celery worker, OCR bounds	Chapters 4, 5, 8, Mathematical Formulations, Schemas
Rasik Dhage	Team Lead, Hosting	Alignment	Frontend-backend integration, GitHub repo, GeM cloud costing	Chapters 1, 2, 7, MeghRaj Costing Model, Roadmap
Dipila Pawar	Frontend Developer	Web application, role-based dashboards, state timeline views	Chapters 2, 3, UI Workflows, Stakeholder Journey	
Vaishnavi Ganjare	Researcher	AICTE APH 2024--27 analysis, *Parshvanath* judicial schedule	Chapters 1, 3, Supreme Court Analysis, CAG Audits	
Shraddha Sabale	PPT	Core Research	System architecture diagrams, impact reduction metrics	Chapters 2, 8, Visual Flowcharts, Feasibility
Janhvi Sapra	CI/CD Manager	Automated pytest harness, Docker setup, DPDP Act 2023 review	Chapters 6, 8, Security Governance, Test Suites	

Team SOUL NETRA 6-Member Engineering Ownership Allocation

Three-Phase Implementation Roadmap

- **Phase 1: Student Prototype & SIH Demonstration (Completed):** Core state machines, 11-table database, split-clock attribution, 12-step jury runner.
- **Phase 2: Regional Bureau Pilot (Months 1--6):** Supervised deployment across 50 technical institutions under Western Regional Office (Mumbai).
- **Phase 3: National Rollout (Months 7--18):** Pan-India rollout across all 8 regional offices and 10,500+ approved institutions on MeghRaj GI Cloud.

Conclusion & Future Scope

Conclusion

SOUL NETRA addresses an urgent, empirical governance challenge in India's technical higher education regulatory administration. By shifting from passive form-intake portals to an explainable, rule-version-aware orchestration and continuing compliance monitoring platform, the system establishes accountability, eliminates bureaucratic finger-pointing through Split-Clock Delay Attribution, and guarantees non-repudiation through a rolling SHA-256 cryptographic audit ledger.

Future Scope

Future enhancements include:

1. Integration with the Ministry of Electronics & IT's **Bhashini API** for automated translation of deficiency notices across 12 scheduled Indian languages.
2. Mobile geo-fenced inspection logging for Expert Visit Committee members conducting on-site laboratory audits.
3. Integration with state land title registries (such as Mahabhulekh in Maharashtra) for real-time verification of 30-year registered land lease renewals.

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Mathematical Notations & Formal State Transitions

Formal Notation Glossary

Symbol / Variable	Formal Definition	Mathematical Domain
$\mathcal{A} = (id, \mathcal{R}, \mathcal{S}, \tau)$	Application regulatory tuple comprising identifier $id \in \mathbb{U}$, route topology $\mathcal{R}$, current stage $\mathcal{S}$, and temporal state vector τ.	
$S(t) \in \Sigma$	Continuous-time stage occupancy function mapping time $t \in [t_0, t_n]$ to the finite set of valid lifecycle stages Σ.	
$T_{\text{inst}}, T_{\text{admin}}, T_{\text{paused}}$	Non-negative real numbers representing accumulated milliseconds under institutional, administrative, or statutory stay custody.	
$\Phi_{\text{inst}}, \Phi_{\text{admin}}$	Percentage proportion of delay attributed to the institution vs. administrative council officers ($\Phi_{\text{inst}} + \Phi_{\text{admin}} = 100$).	
$H_i \in \{0,1\}^{256}$	256-bit cryptographic digest representing block height i in the SHA-256 rolling audit hash chain.	
$\mathcal{D}_{\text{due}} \in \mathbb{T}_{\text{UTC}}$	Derived statutory cut-off timestamp computed by excluding non-working calendar days and state-specific gazetted holidays.	
$\mathbb{G}_{\mathcal{J}} \subset \mathbb{D}$	Finite set of gazetted public holiday dates promulgated for jurisdiction $\mathcal{J}$.	
$\mathcal{O}_{\text{post}} = (id, C, T, \mathcal{E}, \sigma)$	Continuing post-approval obligation tuple with clause citation C, term T, evidence set $\mathcal{E}$, and composite state σ.	

Mathematical Notation & Formal Variable Definitions

Complete Declarative State Transition Matrix

Current State (S_t)	Trigger Event (E)	Transition Guard Condition	Next State (S_{t+1})
`DRAFT`	`SUBMIT_APPLICATION`	All required files uploaded; Fee paid.	`UNDER_SCRUTINY`
`UNDER_SCRUTINY`	`RAISE_DEFICIENCY`	Deficiency items cited to APH clause.	`DEFICIENCY_ISSUED`
`DEFICIENCY_ISSUED`	`SUBMIT_RECTIFICATION`	Institutional response attached before SLA.	`DEFICIENCY_SUBMITTED`
`DEFICIENCY_SUBMITTED`	`ACCEPT_RECTIFICATION`	Scrutiny officer digitally validates proofs.	`SCRUTINY_CLEARED`
`SCRUTINY_CLEARED`	`INITIATE_EVC`	Route in \{ `SCRUTINY_EVC`, `EVC_SAC` \}.	`EVC_SCHEDULED`

Current State (S_t)	Trigger Event (E)	Transition Guard Condition	Next State (S_{t+1})
`EVC_SCHEDULED`	`CONDUCT_EVC`	On-site or digital visit completed by panel.	`EVC_REPORT_SUBMITTED`
`EVC_REPORT_SUBMITTED`	`APPROVE_REPORT`	Regional Director signs off on EVC findings.	`COUNCIL_REVIEW`
`COUNCIL_REVIEW`	`GRANT_APPROVAL`	Final Executive Committee quorum approval.	`APPROVED_ACTIVE`
`COUNCIL_REVIEW`	`REJECT_WITH_SAC`	Formal rejection with appeal window.	`SAC_APPEAL_WINDOW`
`ANY_STAGE`	`JUDICIAL_STAY`	Valid High Court / SC order number logged.	`PAUSED_LITIGATION`

Formal Declarative State Transition Rules & Guard Invariants

Source Code Deposit for Software Copyright Registration

> **Deposit under Rule 70(5) of the Copyright Rules, 2013:** \ In accordance with Rule 70(5) of the Copyright Rules, 2013, governing registration of copyright in computer software works, the following representative source code excerpts comprising key architectural components, relational schemas, state machines, and cryptographic algorithms are deposited as part of the official application.

Module 1: Relational Schema & Audit Trigger DDL (schema.sql)

```
-- SOUL NETRA: Master Database Schema & Tamper-Evident Ledger
-- Engine: PostgreSQL 16 Enterprise | 3NF Relational Integrity

CREATE TABLE institutions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    aicte_permanent_id VARCHAR(50) UNIQUE NOT NULL,
    institution_name TEXT NOT NULL,
    state_jurisdiction VARCHAR(50) NOT NULL,
    regional_office VARCHAR(50) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE applications (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    institution_id UUID NOT NULL REFERENCES institutions(id) ON DELETE RESTRICT,
    application_type VARCHAR(50) NOT NULL,
    route_type VARCHAR(50) NOT NULL CHECK (route_type IN ('ONLY_SCRUTINY', 'SCRUTINY_EVC', 'EVC_SA
C_SHC', 'ONLY_EVC')),
    current_stage VARCHAR(50) NOT NULL DEFAULT 'DRAFT',
    active_clock_owner VARCHAR(20) NOT NULL CHECK (active_clock_owner IN ('INSTITUTION', 'ADMIN',
'PAUSED')),
    institution_clock_ms BIGINT NOT NULL DEFAULT 0,
    admin_clock_ms BIGINT NOT NULL DEFAULT 0,
    paused_clock_ms BIGINT NOT NULL DEFAULT 0,
    stage_entered_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    statutory_deadline TIMESTAMPTZ NOT NULL,
    row_version INTEGER NOT NULL DEFAULT 1,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE TABLE audit_ledger (
    id BIGSERIAL PRIMARY KEY,
    application_id UUID NOT NULL REFERENCES applications(id),
    event_timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    user_id UUID NOT NULL,
    actor_role VARCHAR(50) NOT NULL,
    action_type VARCHAR(50) NOT NULL,
    previous_stage VARCHAR(50),
    new_stage VARCHAR(50),
    previous_hash CHAR(64) NOT NULL,
    current_hash CHAR(64) NOT NULL,
    state_snapshot JSONB NOT NULL
);

-- Immutable Append-Only Ledger Rule: Prevent Retrospective UPDATE or DELETE
CREATE RULE audit_no_update AS ON UPDATE TO audit_ledger DO INSTEAD NOTHING;
CREATE RULE audit_no_delete AS ON DELETE TO audit_ledger DO INSTEAD NOTHING;
```

Module 2: Polymorphic State Machine Engine (workflow_engine.py)

```
# SOUL NETRA: Polymorphic State Machine & Split-Clock Delay Engine
import time
from datetime import datetime, timezone
from typing import Dict, Any, Tuple

class RegulatoryStateMachine:
    ROUTE_TOPOLOGIES = {
        "ONLY_SCRUTINY": [
            ("DRAFT", "UNDER_SCRUTINY"),
            ("UNDER_SCRUTINY", "DEFICIENCY_ISSUED"),
            ("DEFICIENCY_ISSUED", "DEFICIENCY_SUBMITTED"),
            ("DEFICIENCY_SUBMITTED", "SCRUTINY_CLEARED"),
            ("SCRUTINY_CLEARED", "COUNCIL_REVIEW"),
            ("COUNCIL_REVIEW", "APPROVED_ACTIVE"),
        ],
        "SCRUTINY_EVC": [
            ("DRAFT", "UNDER_SCRUTINY"),
            ("UNDER_SCRUTINY", "SCRUTINY_CLEARED"),
            ("SCRUTINY_CLEARED", "EVC_SCHEDULED"),
            ("EVC_SCHEDULED", "EVC_REPORT_SUBMITTED"),
            ("EVC_REPORT_SUBMITTED", "COUNCIL_REVIEW"),
            ("COUNCIL_REVIEW", "APPROVED_ACTIVE"),
        ]
    }

    @classmethod
    def execute_transition(
        cls,
        app: Dict[str, Any],
        target_stage: str,
        user_role: str,
        current_time: datetime
    ) -> Tuple[Dict[str, Any], int, int]:
        """
        Executes atomic state transition with millisecond split-clock accumulation.
        Returns: (updated_app_record, delta_inst_ms, delta_admin_ms)
        """
        route = app["route_type"]
        valid_transitions = cls.ROUTE_TOPOLOGIES.get(route, [])
        if (app["current_stage"], target_stage) not in valid_transitions:
            raise ValueError(f"Illegal transition: {app['current_stage']} -> {target_stage} under
route {route}")

        # Compute elapsed duration since entering current stage
        elapsed_ms = int((current_time - app["stage_entered_at"]).total_seconds() * 1000)
        delta_inst, delta_admin = 0, 0

        if app["active_clock_owner"] == "INSTITUTION":
            delta_inst = elapsed_ms
        elif app["active_clock_owner"] == "ADMIN":
            delta_admin = elapsed_ms

        # Determine next clock owner
        next_clock_owner = "ADMIN" if target_stage in ["UNDER_SCRUTINY", "DEFICIENCY_SUBMITTED",
"COUNCIL_REVIEW"] else "INSTITUTION"

        app["institution_clock_ms"] += delta_inst
        app["admin_clock_ms"] += delta_admin
        app["current_stage"] = target_stage
        app["active_clock_owner"] = next_clock_owner
        app["stage_entered_at"] = current_time
```

```
        app["row_version"] += 1

        return app, delta_inst, delta_admin
```

Module 3: Cryptographic SHA-256 Ledger (audit_ledger.py)

```
# SOUL NETRA: Rolling SHA-256 Hash-Chained Audit Ledger
import hashlib
import json
from datetime import datetime, timezone

def compute_block_hash(prev_hash: str, timestamp: datetime, user_id: str, action: str, snapshot: d
ict) -> str:
    """Computes SHA-256 hash using canonical JSON formatting."""
    canonical_payload = json.dumps(snapshot, sort_keys=True, separators=(',', ':'))
    raw_block = f"{prev_hash}|{timestamp.isoformat()}|{user_id}|{action}|{canonical_payload}"
    return hashlib.sha256(raw_block.encode('utf-8')).hexdigest()

def verify_audit_chain(ledger_records: list) -> Tuple[bool, int]:
    """
    Verifies cryptographic integrity across audit ledger chain.
    Returns: (is_valid, corrupted_block_index)
    """
    current_expected_hash = "0" * 64
    for idx, block in enumerate(ledger_records):
        if block["previous_hash"] != current_expected_hash:
            return False, idx
        recalculated = compute_block_hash(
            block["previous_hash"],
            block["event_timestamp"],
            block["user_id"],
            block["action_type"],
            block["state_snapshot"]
        )
        if recalculated != block["current_hash"]:
            return False, idx
        current_expected_hash = block["current_hash"]
    return True, -1
```